6.8 Interrelationships between Parameters

VTU : June-12, Jan.-05,07,08,09,14,17, July-05,07,08,09,10, Dec.-10,11

We can express any parameter interms of other parameter. In this section we will discuss all such interrelationships between the parameters.

6.8.1 z-Parameters interms of other Parameters

The equations for z-parameters are as follows,

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{... (A)}$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{... (B)}$$

[A] Interms of y-Parameters

The equations for y-parameters are as follows,

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{... (1)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{... (2)}$$

Writing equations in matrix form, we have,

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Solving above equations, using Cramer's rule for V_1 and V_2. We can write,

$$V_1 = \frac{\begin{vmatrix} I_1 & y_{12} \\ I_2 & y_{22} \end{vmatrix}}{\begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}} = \frac{y_{22} I_1 - y_{12} I_2}{y_{11} y_{22} - y_{12} y_{21}}$$

$$= \frac{y_{22}}{y_{11} y_{22} - y_{12} y_{21}} \cdot I_1 - \frac{y_{12}}{y_{11} y_{22} - y_{12} y_{21}} I_2 \quad \text{... (3)}$$

Similarly,

$$V_2 = \frac{\begin{vmatrix} y_{11} & I_1 \\ y_{21} & I_2 \end{vmatrix}}{\begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}} = \frac{y_{11} I_2 - y_{21} I_1}{y_{11} y_{22} - y_{12} y_{21}}$$

$$= \frac{-y_{21}}{y_{11} y_{22} - y_{12} y_{21}} I_1 + \frac{y_{11}}{y_{11} y_{22} - y_{12} y_{21}} I_2 \quad \text{... (4)}$$

Let

$$y_{11} y_{22} - y_{12} y_{21} = \Delta y$$

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Rewriting equations (3) and (4), we have,

$$V_1 = \left[\frac{y_{22}}{\Delta y}\right] I_1 + \left[\frac{-y_{12}}{\Delta y}\right] I_2 \quad \text{... (5)}$$

and

$$V_2 = \left[\frac{-y_{21}}{\Delta y}\right] I_1 + \left[\frac{y_{11}}{\Delta y}\right] I_2 \quad \text{... (6)}$$

Comparing equations (5) and (6) to equations (A) and (B), we have,

$$z_{11} = \frac{y_{22}}{\Delta y} \qquad z_{12} = \frac{-y_{12}}{\Delta y}$$

$$z_{21} = \frac{-y_{21}}{\Delta y} \qquad z_{22} = \frac{y_{11}}{\Delta y}$$

In the matrix form z-parameters can be written as,

$$[z] = \begin{bmatrix} \frac{y_{22}}{\Delta y} & \frac{-y_{12}}{\Delta y} \\ \frac{-y_{21}}{\Delta y} & \frac{y_{11}}{\Delta y} \end{bmatrix}$$

[B] Interms of h-Parameters

The equations for h-parameters are as follows,

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{... (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{... (2)}$$

From equation (2) we can write,

$$h_{22} V_2 = -h_{21} I_1 + I_2$$

$$\therefore \quad V_2 = \left[\frac{-h_{21}}{h_{22}}\right] I_1 + \left[\frac{1}{h_{22}}\right] I_2 \quad \text{... (3)}$$

Substituting value of V_2 in equation (1) we have,

$$V_1 = h_{11} I_1 + h_{12}\left[\frac{-h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2\right]$$

$$\therefore \quad V_1 = \left[h_{11} - \frac{h_{12} h_{21}}{h_{22}}\right] I_1 + \frac{h_{12}}{h_{22}} I_2$$

$$\therefore \quad V_1 = \left[\frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}}\right] I_1 + \left[\frac{h_{12}}{h_{22}}\right] I_2 \quad \text{... (4)}$$

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Comparing equations (4) and (3) with equations (A) and (B) respectively, we have,

$$z_{11} = \frac{h_{11}h_{22} - h_{12}h_{21}}{h_{22}} \qquad z_{12} = \frac{h_{12}}{h_{22}}$$

$$z_{21} = \frac{-h_{21}}{h_{22}} \qquad z_{22} = \frac{1}{h_{22}}$$

In the matrix form z-parameters can be written as,

$$[z] = \begin{bmatrix} \frac{h_{11}h_{22} - h_{12}h_{21}}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

[C] Interms of Transmission (ABCD) Parameters

The equations for transmission parameters are as follows,

$$V_1 = A\,V_2 + B\,(-I_2) \qquad \text{... (1)}$$

$$I_1 = C\,V_2 + D\,(-I_2) \qquad \text{... (2)}$$

We can rewrite equation (2) as follows,

$$C\,V_2 = I_1 + D\,I_2$$

$$\therefore \quad V_2 = \left[\frac{1}{C}\right] I_1 + \left[\frac{D}{C}\right] I_2 \qquad \text{... (3)}$$

Substituting value of V_2 in equation (1), we have,

$$V_1 = A\left[\left(\frac{1}{C}\right) I_1 + \left(\frac{D}{C}\right) I_2\right] - B \cdot I_2$$

$$\therefore \quad V_1 = \left[\frac{A}{C}\right] I_1 + \left[\frac{AD}{C} - B\right] I_2$$

$$\therefore \quad V_1 = \left[\frac{A}{C}\right] I_1 + \left[\frac{AD - BC}{C}\right] I_2 \qquad \text{... (4)}$$

Comparing equations (4) and (3) with the equations (A) and (B) respectively, we have,

$$z_{11} = \frac{A}{C} \qquad z_{12} = \frac{AD - BC}{C}$$

$$z_{21} = \frac{1}{C} \qquad z_{22} = \frac{D}{C}$$

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In the matrix form, the z-parameters can be written as,

$$[z] = \begin{bmatrix} \frac{A}{C} & \frac{AD-BC}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$$

6.8.2 y-Parameters interms of other Parameters

The equations for y-parameters are as follows,

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{... (A)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{... (B)}$$

[A] Interms of z-parameters

The equations for z-parameters are as follows,

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{... (1)}$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{... (2)}$$

Writing equations in matrix form,

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Solving above equations using Cramer's rule for I_1 and I_2, we can write,

$$I_1 = \frac{\begin{vmatrix} V_1 & z_{12} \\ V_2 & z_{22} \end{vmatrix}}{\begin{vmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{vmatrix}} = \frac{z_{22} \cdot V_1 - z_{12} \cdot V_2}{z_{11} \cdot z_{22} - z_{12} \cdot z_{21}}$$

$$= \frac{z_{22}}{z_{11} z_{22} - z_{12} z_{22}} \cdot V_1 + \frac{-z_{12}}{z_{11} z_{22} - z_{12} z_{21}} \cdot V_2 \quad \text{...(3)}$$

Similarly,

$$I_2 = \frac{\begin{vmatrix} z_{11} & V_1 \\ z_{21} & V_2 \end{vmatrix}}{\begin{vmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{vmatrix}} = \frac{z_{11} \cdot V_2 - z_{21} \cdot V_1}{z_{11} \cdot z_{22} - z_{12} \cdot z_{21}}$$

$$= \frac{-z_{21}}{z_{11} \cdot z_{22} - z_{12} z_{21}} \cdot V_1 + \frac{z_{11}}{z_{11} z_{22} - z_{12} z_{21}} \cdot V_2 \quad \text{... (4)}$$

Let $\Delta z = z_{11} \cdot z_{22} - z_{12} \cdot z_{21}$

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Rewriting equations (3) and (4),

$$I_1 = \left[\frac{z_{22}}{\Delta z}\right] V_1 + \left[\frac{-z_{12}}{\Delta z}\right] V_2 \quad \text{... (5)}$$

and

$$I_2 = \left[\frac{-z_{21}}{\Delta z}\right] V_1 + \left[\frac{z_{11}}{\Delta z}\right] V_2 \quad \text{... (6)}$$

Comparing equations (5) and (6) with the equations (A) and (B) respectively, we have,

$$y_{11} = \frac{z_{22}}{\Delta z} \qquad y_{12} = \frac{-z_{12}}{\Delta z}$$

$$y_{21} = \frac{-z_{21}}{\Delta z} \qquad y_{22} = \frac{z_{11}}{\Delta z}$$

In the matrix form, y-parameters can be written as,

$$[y] = \begin{bmatrix} \frac{z_{22}}{\Delta z} & \frac{-z_{12}}{\Delta z} \\ \frac{-z_{21}}{\Delta z} & \frac{z_{11}}{\Delta z} \end{bmatrix}$$

[B] Interms of h-Parameters

The equations for h-parameters are as follows,

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{... (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{... (2)}$$

We can rewrite equation (1) as follows,

$$h_{11} I_1 = V_1 - h_{12} V_2$$

$$\therefore \quad I_1 = \left[\frac{1}{h_{11}}\right] V_1 + \left[\frac{-h_{12}}{h_{11}}\right] V_2 \quad \text{...(3)}$$

Substituting equation (3) in equation (2), we have,

$$I_2 = h_{21}\left[\frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2\right] + h_{22} V_2$$

$$\therefore \quad I_2 = \left[\frac{h_{21}}{h_{11}}\right] V_1 + \left[h_{22} - \frac{h_{12} h_{21}}{h_{11}}\right] V_2$$

$$\therefore \quad I_2 = \left[\frac{h_{21}}{h_{11}}\right] V_1 + \left[\frac{h_{11} h_{22} - h_{12} h_{21}}{h_{11}}\right] V_2 \quad \text{... (4)}$$

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Comparing equations (3) and (4) with the equations (A) and (B) respectively, we have,

$$y_{11} = \frac{1}{h_{11}} \qquad y_{12} = \frac{-h_{12}}{h_{11}}$$

$$y_{21} = \frac{h_{21}}{h_{11}} \qquad y_{22} = \frac{h_{11}h_{22} - h_{12}h_{21}}{h_{11}}$$

In the matrix form y-parameters can be written as,

$$[y] = \begin{bmatrix} \frac{1}{h_{11}} & \frac{-h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{h_{11}h_{22} - h_{12}h_{21}}{h_{11}} \end{bmatrix}$$

[C] Interms of Transmission (ABCD) Parameters

The equations for transmission parameters are as follows,

$$V_1 = A\,V_2 + B\,(-I_2) \qquad \text{... (1)}$$

$$I_1 = C\,V_2 + D\,(-I_2) \qquad \text{... (2)}$$

We can rewrite equation (1) as follows,

$$-B \cdot I_2 = V_1 - A\,V_2$$

$$\therefore \quad I_2 = \left[-\frac{1}{B}\right] V_1 + \left[\frac{A}{B}\right] V_2 \qquad \text{... (3)}$$

Substituting value of I_2 in equation (2), we have,

$$I_1 = C \cdot V_2 + D\left[\frac{+1}{B} V_1 - \frac{A}{B}\right] V_2$$

$$\therefore \quad I_1 = \left[\frac{D}{B}\right] V_1 + \left[C - \frac{AD}{B}\right] V_2$$

$$\therefore \quad I_1 = \left[\frac{D}{B}\right] V_1 + \left[\frac{BC - AD}{B}\right] V_2 \qquad \text{... (4)}$$

Comparing equations (4) and (3) with the equations (A) and (B) respectively, we have,

$$y_{11} = \frac{D}{B} \qquad y_{12} = \frac{BC - AD}{B}$$

$$y_{21} = \frac{-1}{B} \qquad y_{22} = \frac{A}{B}$$

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In the matrix form, y-parameters can be written as,

$$[y] = \begin{bmatrix} \frac{D}{B} & \frac{BC-AD}{B} \\ \frac{-1}{B} & \frac{A}{B} \end{bmatrix}$$

6.8.3 h-Parameters interms of other Parameters

The equations for h-parameters are as follows,

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{... (A)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{... (B)}$$

[A] Interms of z-Parameters

The equations of z-parameters are as follows,

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{... (1)}$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{... (2)}$$

We can rewrite equations (2) as,

$$z_{22} I_2 = -z_{21} I_1 + V_2$$

$$\therefore \quad I_2 = \left[\frac{-z_{21}}{z_{22}}\right] I_1 + \left[\frac{1}{z_{22}}\right] V_2 \quad \text{... (3)}$$

Substituting value of I_2 in equation (1), we have,

$$V_1 = z_{11} I_1 + z_{12}\left(\left[-\frac{z_{21}}{z_{22}}\right] I_1 + \left[\frac{1}{z_{22}}\right] V_2\right)$$

$$\therefore \quad V_1 = \left[z_{11} - \frac{z_{12} z_{21}}{z_{22}}\right] I_1 + \left[\frac{z_{12}}{z_{22}}\right] V_2$$

$$\therefore \quad V_1 = \left[\frac{z_{11} z_{22} - z_{12} z_{21}}{z_{22}}\right] I_1 + \left[\frac{z_{12}}{z_{22}}\right] V_2 \quad \text{... (4)}$$

Comparing equations (4) and (3) with equations (A) and (B) respectively, we have,

$$h_{11} = \frac{z_{11} z_{22} - z_{12} z_{21}}{z_{22}}, \quad h_{12} = \frac{z_{12}}{z_{22}}$$

$$h_{21} = \frac{-z_{21}}{z_{22}}, \quad h_{22} = \frac{1}{z_{22}}$$

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In the matrix form, h-parameters can be written as,

$$[h] = \begin{bmatrix} \dfrac{z_{11}z_{22} - z_{12}z_{21}}{z_{22}} & \dfrac{z_{12}}{z_{22}} \\ \dfrac{-z_{21}}{z_{22}} & \dfrac{1}{z_{22}} \end{bmatrix}$$

[B] Interms of y-Parameters

The equations for y-parameters are as follows,

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{... (1)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{... (2)}$$

We can rewrite equation (1) as follows,

$$y_{11} V_1 = I_1 - y_{12} V_2$$

$$\therefore \quad V_1 = \left[\frac{1}{y_{11}}\right] I_1 + \left[\frac{-y_{12}}{y_{11}}\right] V_2 \quad \text{... (3)}$$

Substituting value of V_1 in equation (3), we have,

$$I_2 = y_{21}\left(\left[\frac{1}{y_{11}}\right] I_1 + \left[\frac{-y_{12}}{y_{11}}\right] V_2\right) + y_{22} V_2$$

$$\therefore \quad I_2 = \left[\frac{y_{21}}{y_{11}}\right] I_1 + \left[y_{22} - \frac{y_{12} y_{21}}{y_{11}}\right] V_2$$

$$\therefore \quad I_2 = \left[\frac{y_{21}}{y_{11}}\right] I_1 + \left[\frac{y_{11} y_{22} - y_{12} y_{21}}{y_{11}}\right] V_2 \quad \text{... (4)}$$

Comparing equations (3) and (4) with equations (A) and (B) respectively, we have,

$$h_{11} = \frac{1}{y_{11}} \qquad h_{12} = \frac{-y_{12}}{y_{11}}$$

$$h_{21} = \frac{y_{21}}{y_{11}} \qquad h_{22} = \frac{y_{11} y_{22} - y_{12} y_{21}}{y_{11}}$$

In the matrix form, h-parameters can be written as,

$$[h] = \begin{bmatrix} \dfrac{1}{y_{11}} & \dfrac{-y_{12}}{y_{11}} \\ \dfrac{y_{21}}{y_{11}} & \dfrac{y_{11} y_{22} - y_{12} y_{21}}{y_{11}} \end{bmatrix}$$

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[C] Interms of Transmission Parameters

The equations for transmission parameters are as follows,

$$V_1 = A V_2 + B(-I_2) \quad \text{... (1)}$$

$$I_1 = C V_2 + D(-I_2) \quad \text{... (2)}$$

We can rewrite equation (2) as follows,

$$-D I_2 = I_1 - C V_2$$

$$\therefore \quad I_2 = \left[\frac{-1}{D}\right] I_1 + \left[+\frac{C}{D}\right] V_2 \quad \text{... (3)}$$

Substituting value of I_2 in equation (1), we have,

$$V_1 = A V_2 + B\left\{-\left(\left[-\frac{1}{D}\right] I_1 + \left[\frac{C}{D} V_2\right]\right)\right\}$$

$$\therefore \quad V_1 = A V_2 + \left[\frac{B}{D}\right] I_1 - \frac{BC}{D} V_2$$

$$\therefore \quad V_1 = \left[\frac{B}{D}\right] I_1 + \left[\frac{AD - BC}{D}\right] V_2 \quad \text{... (4)}$$

Comparing equations (4) and (3) with equations (A) and (B) respectively we have,

$$h_{11} = \frac{B}{D}, \quad h_{12} = \frac{AD - BC}{D}$$

$$h_{21} = -\frac{1}{D}, \quad h_{22} = \frac{C}{D}$$

In the matrix form, h-parameters can be written as,

$$[h] = \begin{bmatrix} \frac{B}{D} & \frac{AD - BC}{D} \\ -\frac{1}{D} & \frac{C}{D} \end{bmatrix}$$

6.8.4 Transmission (ABCD) Parameters Interms of other Parameters

The equations for transmission parameters are as follows,

$$V_1 = A V_2 + B(-I_2) \quad \text{...(A)}$$

$$I_1 = C V_2 + D(-I_2) \quad \text{... (B)}$$

[A] Interms of z-Parameters

The equations for z-parameters are as follows,

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{...(1)}$$

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$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{... (2)}$$

We can rewrite equation (2) as follows,

$$-z_{21} I_1 = -V_2 + z_{22} I_2$$

$$\therefore \quad I_1 = \left[+\frac{1}{z_{21}}\right] V_2 + \left[\frac{-z_{22}}{z_{21}}\right] I_2 \quad \text{...(3)}$$

Substituting value of I_1 in equation (1), we have,

$$V_1 = z_{11}\left[\left(\frac{1}{z_{21}}\right)V_2 + \left(\frac{-z_{22}}{z_{21}}\right)I_2\right] + z_{12} I_2$$

$$\therefore \quad V_1 = \left[\frac{z_{11}}{z_{21}}\right] V_2 + \left[\frac{-z_{11}z_{22}}{z_{21}} + z_{12}\right] I_2$$

$$\therefore \quad V_1 = \left[\frac{z_{11}}{z_{21}}\right] V_2 + \left[\frac{+z_{11}z_{22} - z_{12}z_{21}}{z_{21}}\right](-I_2) \quad \text{...(4)}$$

Comparing equations (4) and (3) with equations (A) and (B) respectively, we have,

$$A = \frac{z_{11}}{z_{21}}, \quad B = \frac{z_{11}z_{22} - z_{21}z_{22}}{z_{21}}$$

$$C = \frac{1}{z_{21}}, \quad D = \frac{z_{22}}{z_{21}}$$

In the matrix form, transmission parameters can be written as,

$$[T] = \begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{z_{11}z_{22} - z_{12}z_{21}}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$$

[B] Interms of y-Parameters

The equations for y-parameters are as follows,

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{...(1)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{...(2)}$$

We can rewrite equation (2) as follows,

$$y_{21} V_1 = -y_{22} V_2 + I_2$$

$$\therefore \quad V_1 = \left[\frac{-y_{22}}{y_{21}}\right] V_2 + \left[\frac{-1}{y_{21}}\right](-I_2) \quad \text{...(3)}$$

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Substituting value of V_1 in equation (1), we have,

$$I_1 = y_{11}\left[\frac{-y_{22}}{y_{21}}V_2 + \frac{1}{y_{21}}I_2\right] + y_{12}V_2$$

$$\therefore \quad I_1 = \left[y_{12} - \frac{y_{11}y_{22}}{y_{21}}\right]V_2 + \left[\frac{-1}{y_{21}}\right](-I_2)$$

$$\therefore \quad I_1 = \left[\frac{y_{12}y_{21} - y_{11}y_{22}}{y_{21}}\right]V_2 + \left[\frac{-y_{11}}{y_{21}}\right](-I_2) \quad ...(4)$$

Comparing equations (3) and (4) with the equations (A) and (B), we have,

$$A = \frac{-y_{22}}{y_{21}} \qquad B = \frac{-1}{y_{21}}$$

$$C = \frac{y_{12}y_{21} - y_{11}y_{22}}{y_{21}} \qquad D = \frac{-y_{11}}{y_{21}}$$

In the matrix form, transmission parameters can be written as,

$$[T] = \begin{bmatrix} \frac{-y_{22}}{y_{21}} & \frac{-1}{y_{21}} \\ \frac{y_{12}y_{21} - y_{11}y_{22}}{y_{21}} & \frac{-y_{11}}{y_{21}} \end{bmatrix}$$

[C] Interms of h-Parameters

The equations for h-parameters are as follows,

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad ...(1)$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad ...(2)$$

We can rewrite equation (2) as follows,

$$h_{21}I_1 = I_2 - h_{22}V_2$$

$$\therefore \quad I_1 = \left[\frac{-h_{22}}{h_{21}}\right]V_2 + \left[\frac{-1}{h_{21}}\right](-I_2) \quad ...(3)$$

Substituting value of I_1 in equation (1) we have,

$$V_1 = h_{11}\left[\frac{-h_{22}}{h_{21}}V_2 + \frac{1}{h_{21}}I_2\right] + h_{12}V_2$$

$$\therefore \quad V_1 = \left[\frac{h_{12}h_{21} - h_{11}h_{22}}{h_{21}}\right]V_2 + \left[\frac{-h_{11}}{h_{21}}\right](-I_2) \quad ...(4)$$

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Comparing equations (4) and (3) with the equations (A) and (B) respectively, we have,

$$A = \frac{h_{12}h_{21} - h_{11}h_{22}}{h_{21}} \qquad B = \frac{-h_{11}}{h_{21}}$$

$$C = \frac{-h_{22}}{h_{21}} \qquad D = \frac{-1}{h_{21}}$$

In the matrix form, transmission parameters can be written as,

$$[T] = \begin{bmatrix} \frac{h_{12}h_{21} - h_{11}h_{22}}{h_{21}} & \frac{-h_{11}}{h_{21}} \\ \frac{-h_{22}}{h_{21}} & \frac{-1}{h_{21}} \end{bmatrix}$$

Table 6.8.1 gives interrelationships between all the parameters.

	[z]	[y]	[h]	[T]
[z]	$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{y_{22}}{\Delta y} & \frac{-y_{12}}{\Delta y} \\ \frac{-y_{21}}{\Delta y} & \frac{y_{11}}{\Delta y} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$
[y]	$\begin{bmatrix} \frac{z_{22}}{\Delta z} & \frac{-z_{12}}{\Delta z} \\ \frac{-z_{21}}{\Delta z} & \frac{z_{11}}{\Delta z} \end{bmatrix}$	$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{h_{11}} & \frac{-h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{D}{B} & \frac{\Delta T}{B} \\ \frac{1}{B} & \frac{A}{B} \end{bmatrix}$
[h]	$\begin{bmatrix} \frac{\Delta z}{z_{22}} & \frac{z_{21}}{z_{22}} \\ \frac{-z_{21}}{z_{22}} & \frac{1}{z_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{y_{11}} & \frac{-y_{12}}{y_{11}} \\ \frac{y_{21}}{y_{11}} & \frac{\Delta y}{y_{11}} \end{bmatrix}$	$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ \frac{1}{D} & \frac{C}{D} \end{bmatrix}$
[T]	$\begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\Delta z}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$	$\begin{bmatrix} \frac{y_{22}}{y_{21}} & \frac{-1}{y_{21}} \\ \frac{-\Delta y}{y_{21}} & \frac{y_{11}}{y_{21}} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta h}{h_{21}} & \frac{h_{11}}{h_{21}} \\ \frac{h_{22}}{h_{21}} & \frac{1}{h_{21}} \end{bmatrix}$	$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$

Table 6.8.1 Interrelation between various parameters

$\Delta z = z_{11} z_{22} - z_{12} z_{21}$, $\quad \Delta y = y_{11}y_{22} - y_{12} y_{21}$

$\Delta h = h_{11} h_{22} - h_{12} h_{21}$, $\quad \Delta T = AD - BC$

Example 6.8.1 *The z-parameters of a two port network are* $z_{11} = 20\ \Omega$, $z_{22} = 30\ \Omega$, $z_{12} = z_{21} = 10\ \Omega$.

Find y and ABCD parameters of the network. **June-12, Marks 6**

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